Introduction

Mental health and rehabilitation are significant challenges for healthcare systems today. The World Health Organization (WHO) estimates that about one in seven people worldwide lives with a mental health condition. Despite the prevalence of these issues, accessing effective and personalised treatment remains limited, especially in developing countries like South Africa. The South African Depression and Anxiety Group (SADAG) highlights this gap, reporting fewer than one psychiatrist per 100,000 people in the public sector. This creates immense pressure on rehabilitation centres and therapy services.

The purpose of this project is to develop a system that predicts whether a patient is more introverted or extroverted based on behavioural data collected during intake. By providing a data-driven recommendation, the goal is to help clinicians better match patients with the therapy programme that best supports their engagement and progress.

1. Problem Statement

A key concept in psychology that influences therapy effectiveness is the introversion-extroversion personality dimension. Research shows that people respond differently to treatment based on their personality. Extroverted individuals often thrive in group therapy, social exposure exercises, and interactive rehabilitation programs. In contrast, introverted individuals generally benefit more from one-on-one sessions, reflective activities like journaling, and structured counselling. When therapy doesn't align with a person's personality, they are less likely to stay engaged, progress tends to slow, and dropout rates increase.

1.1 The Real-World Problem

Although the link between personality type and therapy effectiveness is recognised, many rehabilitation centres lack an easy and reliable way to assess a patient's personality upon arrival. Tools like the Myers-Briggs Type Indicator and the Big Five Inventory exist, but they require time to complete and need trained professionals. In under-resourced settings, using these tools isn't practical, leading to their frequent omission. Consequently, therapists often rely on quick observations, instincts, or what patients disclose about themselves, which may not always be accurate.

This creates a serious issue. Many patients find themselves in therapy programs that don't really suit their personalities. This mismatch reduces their likelihood of full engagement, slows their progress, and some eventually drop out of treatment altogether. For rehabilitation centers already operating at capacity, this issue wastes valuable time and resources.

The repercussions affect everyone involved. Patients may not receive the necessary level of care and could take longer to recover. Therapists and counselors often have to revisit and adjust treatment plans once they recognise an approach isn't effective. Rehabilitation centers face higher dropout rates and poorer overall outcomes. On a larger scale, healthcare funders and public systems end up financing treatment cycles that are less effective than they should be.

1.2 Proposed Data Science Solution

This project aims to develop a Personality-Based Therapy Allocation System, a machine learning model that helps clinicians make better decisions from the start of treatment. Instead of relying on lengthy assessments or guesswork, the system predicts whether a rehabilitation patient is more introverted or extroverted based on simple behavioural data collected during intake. This may include factors like how much time a person spends alone, how often they engage socially, their comfort level in public settings, and the sise of their social circle.

Based on this prediction, the system then suggests a therapy approach that would likely suit the patient. The goal is not to replace the clinician, but to provide a clear, data-driven recommendation to better match patients with the care they need to engage, progress, and recover more effectively.

The solution will be implemented through the following components:

- A structured MySQL relational database to securely store and manage patient behavioural intake data.

- Exploratory Data Analysis in Python to identify patterns and behavioural trends across personality types.

- A machine learning classification model trained to predict patient personality type (Introvert or Extrovert) from intake behavioural data.

- A PowerBI dashboard presenting patient personality distributions, therapy allocation insights, and model performance metrics for rehabilitation centre management.

1.3 Problem Type

This project represents a binary classification problem. The model aims to predict personality type, which has two possible categories: introvert or extrovert. To achieve this, the model is trained with labelled historical data, meaning data from individuals with known personality types. By learning from this existing data, it can make predictions about new patients during the intake process.

Classification is appropriate here because the goal is to assign each patient to one of two distinct groups. Since the outcome is categorical and limited to two options, the model's task is to make the most accurate assignment possible to help guide clinicians toward better therapy decisions.

1.4 Why Data Science is the Right Approach

Traditional personality assessments may be clinically valid, but they often lack practicality in busy or under-resourced rehabilitation settings. They take time, require trained professionals, and rely on patients' engagement during longer sessions. Many centres simply don't have the capacity for this. A machine learning model provides a more practical alternative. It generates personality predictions in seconds using a short set of behavioural questions, making it far more suitable for routine clinical use.

Using data science in this context also offers important benefits. It allows the system to learn from large amounts of patient data, improving accuracy over time. It reduces reliance on subjective judgment, helping to minimise bias in identifying personality. Instead of guesswork, clinicians receive recommendations based on clear, measurable patterns in the data. The system can handle many patients simultaneously without overburdening staff, which is especially vital in high-demand environments. Additionally, the results can be displayed visually, enabling even non-technical healthcare professionals to easily understand and apply the insights in their decision-making.

1.5 Who This Helps

The primary beneficiaries of this system are:

- **Rehabilitation Centers and Therapy Facilities** — They can use the system to allocate patients to appropriate therapy programs upon intake, reducing mismatches and improving overall recovery outcomes.

- **Therapists and Counselors** — They receive data-driven personality insights at the start of treatment, allowing them to customise their approach and reduce wasted sessions.

- **Patients** — They benefit from therapy programs that better match how they engage with the world, leading to greater comfort, participation, and faster recovery.

- **Public Health Systems** — They see the benefits of reduced treatment costs, lower dropout rates, and more efficient use of limited mental health resources.

In summary, this project addresses a clearly defined and clinically significant real-world problem. Using machine learning to predict patient personality type from behavioural data, it offers rehabilitation centres a practical, scalable tool to improve therapy allocation, boost patient engagement, and support better mental health outcomes, all grounded in solid data science methodologies.

2 Dataset Description

2.1 Data Source and History

The dataset used in the current project was taken from the Myers-Briggs Personality Type Dataset (MBTI). It can be accessed via Kaggle and is made up of posts from PersonalityCafe – a large community where members discuss personality theories and share their MBTI types. The main purpose of collecting such a data set was to aid natural language processing research on expression of different personality types in writing. The data set consists of 8675 entries, where each row represents information about one particular user's personality type and their last 50 posts made on the PersonalityCafe forum.

2.2 Description of Variables

There are two major variables in the dataset which form the basis for the Personality-Based Therapy Allocation System:

- **Type (Target Variable):** Categorical variable which indicates the type of the person's Myers-Briggs Type (for instance, INFJ, ENFP). For this project, these 16 possible personality types were simplified into two dimensions: Introversion (I) vs. Extroversion (E).

- **Posts (Feature Variable):** Raw text of the last 50 posts made by each user.

2.3 Size and Suitability

The current data set is quite large with 8675 unique rows. However, its size is appropriate for the problem at hand for the following reasons:

- **Sufficient Number of Examples:** The number of samples for each type is enough to train a reliable machine learning algorithm that can recognise patterns in people's language according to their personality types.

- **Applicability to a Real-life Problem:** Posts variable stands as a digital equivalent of behavioural intake data mentioned in the Problem Statement. A person's linguistic expression allows one to make predictions even when traditional clinical methods are difficult to apply due to resource limitations.

- **Relational Structure:** The dataset structure allows for building a relational database in MySQL, including tables with personality type definitions and buzzwords.

3. Database Design and SQL

As far as the practical application of this project is concerned, we created a hypothetical scenario where a mental wellness institute stores patient's forum posts in a MySQL database. For such use cases, our system can help organise and analyse text data so that personality types like introverts and extroverts can be easily recognised. Depending on their personality type, a mental wellness institute will be able to conduct therapy sessions personalised to their needs. This way, more effective therapy will lead to improved treatment outcomes and recovery time for all patients.

3.1 Data Source and Relation to Therapy

The dataset for this project includes records that include the last 50 posts from an online forum made by people. This dataset was chosen because individuals tend to be natural and authentic while speaking on such forums as well as when writing down their thoughts and problems in therapy journals. Such "natural language" is a crucial component of this model. If this model can interpret the natural language of patients who write their honest feelings on forums, then it can be extremely effective in real therapy session scenarios.

3.2 Automated Data Insertion/Ingestion (Python to MySQL)

I created a database for this project, named as therapy_db. Where we would be working in. Then, Data ingestion was done using Python to push the data/table to the database. After successfully pushing the data from Python to the database, everything else was done via MySQL.

For ingestion purposes, I used the df.to_sql() function with if_exists='replace' in Python. The patient_record table/dataframe was already cleaned before ingestion

3.3 Detailed Table Explanations

In order to structure chaotic text data, a relational schema consisting of five major tables was devised, each having its own purpose in clinical analysis:

patient_records (main table)

- **Purpose:** This main table will hold all data. The patient_id serves as the Primary Key, ensuring uniqueness of each patient.

- **Utility:** It acts as the “Source of Truth.” Storing posts and personality codes here connects all further analysis to a specific person.

personality_definitions (context provider)

- **Purpose:** Lookup table holding the full names and definitions for each personality code/type.

- **Utility:** Solves the “Readability Problem” by giving clinicians a chance to see the whole definition. Without the need to interpret what "I" or "E" means, a therapist can instantly get all necessary psychological data on the patient.

Different_personalities (census table)

- **Purpose:** Summary table counting how many patients belong to each personality type.

- **Utility:** Helps to allocate resources. If, for example, 80% of people belong to the Introvert group, the institution should hire therapists who know how to communicate with these kind of patients.

personality_characters (complexity tracker)

- **Purpose:** This table performs analysis of post length and complexity in terms of average/min/max characters.

- **Utility:** Helps to understand a patients writing style and communication needs. Very low average characters per post means problems with expressing oneself, therefore the therapist should try to find a different way to make patients talk about their issues.

personality_buzzwords (risk screener)

- **Purpose:** A table counting the number of emotionally dangerous words ("depressed," "help" and "suicide") in each of the two personality types' posts.

- **Utility:** The most important element of this is that it ensures the safety of patients because of its ability to spot high-risk patients and take measures immediately.

3.4 SQL Queries Used to Analyse the Dataset

The power of our database to analyse text lies in queries:

- **Aggregate Queries:** Used COUNT and GROUP BY functions in order to describe the distribution among the two types.

- **Linguistic analysis:** Used LENGTH() to measure character count in posts and rounding with ROUND(). It provides a metric of patient's psychological state.

- **Relational joins:** Used INNER JOIN to combine the tables patient_records and personality_definitions in order to create a human-readable report listing patient's id, their full personality name and their posts.

- **Relational integrity:** Created a Foreign Key (fk_personality_type) to link the two tables (patient_records and personality_definitions) to make sure that the patient is assigned only to a personality defined in the personality_definitions table.

3.5 Idempotent Design for Reliable Operations

This SQL script has been constructed with an idempotent property, which means that it can be run multiple times without causing errors or duplicating records. The technique employed is called "drop for idempotency." Before creating a new table or adding columns to a table, the system deletes any previous version of the table or column from the database.

Reason for this: In case the SQL code is run multiple times, such as when the code is tested several times by an instructor or group member, the MySQL database will update the tables each time without throwing any error messages about "Duplicate Column."

4. Exploratory Data Analysis (EDA) and Visualisations

4.1 Introduction

Before implementing any type of machine learning algorithm, it was wise to first analyse the data itself and comprehend what the data really looked like. This part of the analysis will cover the steps of data exploration on the MBTI Personality Type dataset. The main aim of this task was to identify any patterns between the writing styles of Introverts and Extroverts. All of this analysis was done in Python, utilising a few packages (pandas, matplotlib, and seaborn), resulting in a total of ten visualisations in PNG format.

4.2 Loading the Data and Creating New Features

The cleaned dataset was loaded from a file called cleaned_data.csv. Only two columns were needed the actual post text and the personality type. From there, three new features were created to help measure how people write:

- **word_count** — how many words are in each post
- **char_count** — how many characters are in each post
- **avg_word_length** — the average length of the words used

These three features make it easier to compare writing styles in a straightforward, measurable way.

4.3 The Visualisations and What They Show

Visualisation 1: Number of Introverts vs Extroverts (Bar Chart)

This simple visualisation displays the number of people who belong to the category of either Introverts or Extroverts. It is evident from the graph that there is an imbalanced distribution of introverts and extroverts in the dataset. This will be significant while training our classification model in later stages.

Visualisation 2: Number of Words Per Post (Bar Chart)

This bar graph compares the average number of words used in each post by members of both categories. For Introverts, the average word count was 1,264.6, whereas it was 1,261.2 for Extroverts. The difference in word counts between both categories is only 3 words. Therefore, this factor does not appear to be effective in separating the classes based on their post length.

Visualisation 3: Distribution of Introverts vs Extroverts (Pie Chart)

The class imbalance can easily be identified from the pie chart. The proportion of Introverts in the dataset is 77%, while the proportion of Extroverts is just 23%. The class distribution is clearly highly imbalanced in nature.

Visualisation 4: Distribution of Word Counts by Personality Types (Box Plot)

Not only the averages but also the complete spread of word counts are illustrated here for each personality type. While both groups have nearly identical medians of 1,300 words per post, the spread of the Introverts' data set is slightly narrower, ranging from about 1,100 to 1,500 words. Conversely, the Extroverts' data has a wider box, implying a greater variance among the group members. Also, both distributions include outliers with extremely low numbers of words used in some posts.

Visualisation 5: Word Count Distribution (Overlapping Histogram with Kernel Density Estimate)

This plot provides an opportunity to overlay both personality types and observe their word count distributions. The shapes of the two histograms appear quite similar – each shows a normal distribution peaking in the same 1,300 to 1,500 words. The height of the Introverts' histogram is higher, which means the number of Introvert samples in the dataset is greater than that of the Extroverts'**.**

Visualisation 6: Average Number of Characters Per Post by Personality Type (Bar Graph)

The above chart examines the total number of characters per post instead of the word count. On the whole, Extroverts have an average of 6,905 characters per post, just slightly higher than that of Introverts with 6,893. The difference here is even less pronounced than in the case of the word count with just 12 characters' difference. It seems, therefore, that although Extroverts use somewhat more characters, on average, the length of their posts is virtually equal to that of Introverts'.

Visualisation 7: Average Number of Characters Per Word by Personality Type (Bar Graph)

In this last visualisation, we look at how long the words are used by each personality type. Extroverts use words averaging 4.435 characters in length, whereas Introverts use words averaging 4.405 characters in length. Even though the difference is not too large, it supports the results concerning the number of characters used per post. A consistent However, even a small difference of this sort can be helpful for the algorithm.

Visualisation 8: Average Sentence Length by Personality Type (Bar Graph)

The graph shows the average sentence length based on punctuation, including full stops, exclamations, and questions. The average sentence length of Introverts was 6.06 words, while that of Extroverts was 2.83 words. This is quite a difference, implying that

Introverts in this dataset are likely to have more complicated sentences than their counterparts.

Visualisation 9: Vocabulary Variety by Personality Type (Bar Graph)

This figure reflects the vocabulary variety of each personality type in the dataset. It was measured by the unique word ratio, which refers to the proportion of unique words among all words used. Introverts' unique word ratio was 1.0000, while Extroverts' unique word ratio was 0.9167. A high ratio implies a low repetition rate and vice versa.

4.4 What the EDA Showed Overall

In all 9 graphs, the general impression is that there is very little difference between how Extroverts and Introverts communicate using length and word characteristics. All values – average number of words (3.4), average number of characters (12), average word length (0.03) – differ only slightly between groups, while the newly created graphs for the sentence length and vocabulary diversity provide yet another dimension of difference. In this case, however, it can be stated with confidence that the trends are consistent: Extroverts use longer words and have longer sentences on average, but the fact that Introverts are represented by an overwhelming majority in the dataset (77%) should be taken into account. The best use of the visualisations in the current study would likely come from using all three features in conjunction rather than individually, while the significant class imbalance observed in Visualisations 1 and 3 must be solved during model training..

5 Machine Learning Model

5.1 Data Preparation and Cleaning

The initial data was subjected to extensive preprocessing and cleaning. This step is crucial since "noisy" data can confuse machine learning algorithms and lead to poor predictions.

Noise Removal: All links (URLs) and the pipe character (|||), used to separate posts, were removed from the text. This step prevents the model from attempting to learn patterns in irrelevant punctuation and website addresses.

Simplification of the Target Variable: There were 16 distinct Myers-Briggs Type Indicator (MBTI) personality types in the dataset. To simplify the classification problem, we consolidated it into a binary system – Introvert (I) and Extrovert (E).

Exclusion of Cheat Words: Explicit references to any of the 16 personalities or the terms "introverted" and “extroverted” were eliminated from posts. Otherwise, the model could "cheat" and identify personality by simply finding the corresponding label instead of discerning linguistic patterns.

Once pre-processed, the simplified data was exported as cleaned_data.csv and passed on for Exploratory Data Analysis (EDA) and creating visualisations.

5.2 Feature Engineering: TF-IDF Vectorisation

Machine learning models cannot comprehend text but work exclusively with numerical data. As such, we had to introduce a feature vectoriser.

The Approach: TF-IDF measures the importance of a word for a particular post relative to the whole dataset. Thus, it creates a matrix with percentages for each word's importance (essentially, a word weighting system).

Why it Works for Text Data: If a word occurs often in one post but not elsewhere, it receives a high importance rating (being a distinctive feature). In contrast, English stopwords ("the", "and", "is") were excluded from the model.

Feature Limitation: The vectoriser was limited to 5,000 features to prevent the model from being overly distracted by rare vocabulary terms.

5.3 Data Splitting: Training and Testing Sets

Prior to deployment, the data was separated into two distinct parts:

Training Data (80%): This is essentially the textbook for the model. The model analyses the TF-IDF word matrix along with the corresponding personality labels to discover patterns.

Testing Data (20%): This serves as the ultimate test. The model must predict personality types in this data without seeing the answers, proving whether it has learned something truly useful.

5.4 Choice of Model and Handling of Class Imbalance

Logistic Regression was used as the classifier here due to its superior performance in binary classification tasks and ability to handle extremely large and sparse matrices typical for text and TF-IDF representation.

Class Imbalance Issue:

One of the main issues faced was the strong class imbalance: about 80% of all entries belong to Introverts. Therefore, in its initial stage, the algorithm had figured out that guessing "Introvert" in all cases would give it the highest accuracy score.

How the Problem Was Resolved:

In order to fix this problem, class_weight="balanced" was used. This way, the model was forced to equally treat both types equally and hunt for extrovert language cues in the same way it hunted for introvert ones.

Positive Side: As expected, the model was capable of identifying many more extroverts, thus raising Recall for the extrovert class.

Negative Side: In doing so, it was more likely to have false positives and identify introverts as extroverts (thus reducing Precision), resulting in an ever so slightly decrease in accuracy.

This is an inevitable trade-off which had to be made to make sure that data from both classes is processed evenly.

5.5 Real-World Application: User Testing

To connect the model to its practical psychological/therapy context, a testing prompt was prepared. It asks a user the following question: *After a long, stressful week, how do you prefer to spend your Saturday evening to feel like 'yourself' again, and why?*

Based on this input, the model processes this data according to its TF-IDF parameters and outputs the prediction as either an Introvert or an Extrovert (with an accompanying certainty score).

6 Power BI Dashboard

Explanation of Power BI

An interactive dashboard for the personality prediction project was created in Power BI to present data insights in a clear, simple visual format. The dashboard was divided into four pages.

Page 1: Personality Architecture & Distribution

Goal: To create a higher-order overview of the patient community's personality composition.

Chord Diagram (Custom Visual):

- **Description:** Illustrates how the broad trait categories of I/E interact with each other and with the 16 detailed sub-classes.
- **Benefit:** This is a valuable network perspective. It shows the connections between various patient types, allowing facility managers to form cohesive group therapy sessions.

Tree Map:

- **Description:** Rectangles representing the relative number of each distinct personality type present in the facility.
- **Benefit:** Serves as a census device. Administrators are able to immediately spot the personalities represented the most, allowing them to ensure that the facility's activities cater to the majority.

Page 2: Linguistic Complexity & Engagement

Goal: To gauge the structural differences in patient communication depending on personality types.

Box & Whisker Plot (Custom Visual):

- **Description:** Plots distributions, medians, and outliers regarding word counts among the two distinct types.
- **Benefit:** Critical diagnostic tool. Outliers (also known as whiskers) indicate when certain patients use significantly more or fewer words compared to others. This difference can often be a marker for manic/depressive behaviour, withdrawal, or emotional instability.

Bar Graph (Average Word Count):

- **Description:** Compares average word usage among personalities.
- **Benefit:** Sets up a baseline for patient communication. The knowledge that certain patients tend to be more verbose will make it easier to note any drop in engagement levels in relation to the personality profile.

Page 3: Behavioural Marker Identification (Buzzwords)

Goal: To find and analyse common behavioural triggers used in communication.

Word Cloud (Custom Visual):

- **Description:** Visualised cluster where the word size is directly related to its occurrence in the dataset.
- **Benefit:** Offers an instantaneous vibe check. It allows for the immediate detection of major behavioural themes or topics (e.g., "alone," "social," or "energy"), without having to go through thousands of individual entries.

Clustered Column Graph (Buzzword Frequency):

- **Description:** Direct comparison of selected buzzword frequencies in I/E groups.
- **Benefit:** Allows for the identification of specific behavioural patterns. For example, the usage of certain buzzwords may be higher among Introverts. This will allow therapists to incorporate that into their vocabulary and quickly build rapport.

Pie Graph (Buzzword Distribution):

- **Description:** Shows the proportional representation of different keywords in the dataset.
- **Benefit:** Aids in determining what behaviours are universal across all personalities, and which ones are specific to a specific personality.

Page 4: Diagnostics of the Model & Metrics to Assess Trustworthiness

Goal: To assess the accuracy of the model as an instrument of decision making in the clinical setting.

Indicator Gauges (Accuracy & Weighted F1-Score):

- **Description:** Speedometer-style gauge displays performance of the model based on the preset goal (80%).
- **Benefit:** Enables Safety Transparency. This means that medical directors get clear insight into the level of confidence they may have when using the tool in the decision-making process, making sure AI serves only as a co-pilot and not the decision-maker itself.

Metrics Matrix (Heatmap):

- **Description:** Color-coded representation of Precision, Recall, and F1 score of each class.
- **Benefit:** Identifies the Confidence Zone. The staff is aware of the fact that AI is highly efficient when identifying patients belonging to the class of introverts, while in case of extroverts there might be the necessity to monitor its activity more attentively due to possible mistakes.

Precision vs. Recall Clustering Bar Graph:

- **Description:** Illustrates the difference between precision and recall for the two distinct groups
- **Benefit:** Critical for risk analysis. This graph allows medical teams to calculate the probability of False Negative outcome (missed patient) and take further actions.

7 Conclusion:

At its heart, **my project** is about more than just algorithms; it's about making sure people feel seen and supported when they are at their most vulnerable. By moving away from "one-size-fits-all" therapy and towards a data-driven understanding of personality, **I can ensure** that an introvert isn't forced into an overwhelming social setting, and an extrovert isn't left feeling isolated in solo reflection.

Key Takeaways from the Project

- **A Practical Solution for Real-World Pressure:** In environments where psychiatrists are stretched thin, **I have developed** this system to act as a "clinical co-pilot." It bypasses the need for lengthy, expensive assessments by using natural linguistic patterns the way people speak and write to predict their needs in seconds.
- **The Technical Journey:**

    Data Integrity: I transitioned from raw, "noisy" forum data into a structured MySQL environment, ensuring that patient records are secure, searchable, and organized.

    Linguistic Insights: Through my Python-based EDA, I discovered that while word counts are similar across the board, the true lies in sentence complexity and vocabulary variety.

    Smart Modelling: By using Logistic Regression and adjusting for the heavy "Introvert" bias in the data (77%), I created a model that doesn't just guess the majority but actively hunts for "Extrovert" cues to ensure everyone is represented fairly.

- **Visualizing Progress:** My Power BI dashboard turns abstract numbers into actionable insights. It allows clinic managers to see at a glance if their facility has enough group-therapy resources or if a patient's sudden drop in word count might be a red flag for their mental well-being.

The Big Picture

By integrating SQL, Machine Learning, and Interactive Analytics, I have built a tool that supports everyone in the healthcare chain. Patients get faster, more comfortable recoveries; therapists waste less time on mismatched approaches; and public health systems won't stretch their limited resources further. My project proves that while data is cold, the insights I provide can be incredibly human.

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